RITIK RAMESHWAR PATEL

Tempe, Arizona | ritikpatel@asu.edu | (480) 803-1091 | LinkedIn | GitHub | Portfolio | *Available full-time • immediate start • F-1 • CPT/OPT eligible*

WORK EXPERIENCE

Research Assistant | Arizona State University – Speech/Language Genetics Lab – Tempe, AZ | May 2026 – Present

- Engineered classification and regression features from speech, accuracy and adaptive difficulty signals; Python ETL into PostgreSQL (AWS RDS) with feature stores for clustering and recommendation models (Scikit-learn, PyTorch).
- Developing clustering (K-means/DBSCAN), difficulty recommendation and performance prediction models with error analysis dashboards tracking precision, recall and cohort level drift on prototype sessions.

Graduate Teaching Assistant | Arizona State University – W. P. Carey School of Business – Tempe, AZ | Jan 2025 – May 2026

Courses: Analytics for Unstructured Data; ML in Business; Programming for AI and Data Analytics in Business

- Graduate TA for 3 courses (~150 students/course, 20 hrs/week); graded projects on deep learning, LLM fine-tuning, RAG, agentic workflows, embeddings, prompt evaluation, vector databases and model evaluation (faithfulness, retrieval quality).

Data Analyst | WeAgile Software Solutions – Pune, India | Mar 2022 – Jun 2024

Clients: Meta, Microsoft, The Home Depot, Ralph Lauren, Wayfair, Waste Management

- Owned the full ML lifecycle for production computer vision systems – training data ingestion (CVAT, AWS S3), SageMaker distributed training, offline evaluation, Flask model serving, production monitoring for object detection and tracking (YOLO v7, Faster R-CNN, SORT) on 20+ cameras/day (~18-20 hrs footage) across enterprise retail clients.
- Built training data pipelines with OpenCV augmentation, cross validation by site and feature engineering on detection metadata; improved mAP from 0.71 to 0.84 via hyperparameter tuning on Amazon SageMaker (TensorFlow).
- Optimized production detection quality: 38% false positive reduction on a Home Depot self-checkout deployment at matched recall; sustained ~12 FPS GPU inference and batch scoring throughput for nightly event pipelines.
- Implemented model monitoring and drift analysis – tracking precision and recall shifts, data drift on new camera angles and concept drift after store layout changes; triggered retraining when validation slices degraded across locations.
- Packaged inference with versioned ONNX artifacts on S3, prediction logging in MySQL and A/B rollouts of new checkpoints with confusion matrix and error bucket reviews before client release.

Trainee Software Engineer | WeAgile Software Solutions – Pune, India | Nov 2021 – Feb 2022

- Produced ~500 CVAT labeled images/day for object detection training (WM, Wayfair, Irvine Spectrum Center) with labeling QA and inter-rater checks to improve downstream mAP.

SKILLS

- **Machine Learning:** Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, Classification, Regression, Clustering, Model Selection, Error Analysis, Transfer Learning, Cross validation, Feature Engineering, Hyperparameter Tuning, Feature Stores, Random Forest, Gradient Boosting, Ensembling, GPU Training (CUDA)
- **NLP, LLMs and GenAI:** NLP, LLMs, BERT, spaCy, Retrieval Augmented Generation (RAG), LangChain, Hugging Face Transformers, LLM Fine Tuning (LoRA), SentenceTransformers, Embeddings, Prompt Engineering, AI Agents, Multi-Agent Systems, Tool/Function Calling, OpenAI API, Vector Databases (FAISS), RAGAS, Evaluation
- **Computer Vision and Multimodal ML:** Computer Vision, OpenCV, Ultralytics YOLO, Faster R-CNN, SORT, ResNet-18, CNNs, Transformers, Object Detection, Multi Object Tracking, Multimodal Machine Learning, mAP, CVAT
- **MLOps and Deployment:** MLOps, FastAPI, Flask, Docker, MLflow, Model Deployment, Model Serving, Model Monitoring, Drift Detection (Data and Concept Drift), Distributed Training, Experiment Tracking, ETL Pipelines, ONNX, Git, DVC
- **Cloud and Data:** AWS (S3, SageMaker, RDS), SQL, PostgreSQL, MySQL

EDUCATION

Master of Science in Data Science, Analytics and Engineering | May 2026
Arizona State University – Tempe, AZ
Relevant Coursework: Deep Learning, Computer Vision, Statistical Machine Learning, NLP, LLMs, Optimization, Data Mining

Master of Computer Applications (MCA) | May 2022
Savitribai Phule Pune University – Pune, India

RESEARCH AND PROJECTS

Multimodal Wildfire Detection with Social Media Signals | Jan 2026 – May 2026

- Built multimodal ML combining ResNet-18 wildfire imagery (transfer learning) with Reddit text (BERT classifier, spaCy NER); reported precision, recall, F1 and ROC-AUC with cross-validation and error analysis.
- Designed fusion ETL pipelines and geospatial event schema; multimodal alerts fire only when vision and text signals agree – README includes architecture diagram and metrics.

Production ML Model Deployment | 2026

- Deployed PyTorch CV model as FastAPI + Docker service; MLflow tracking, model versioning, S3 artifacts – documented p95 inference latency, batch throughput and rollback steps.
- Implemented an end-to-end train, evaluate, serve lifecycle on GitHub: architecture diagram, deployment instructions, screenshots and monitoring hooks for production ML.

LLM RAG Application with Evaluation | 2026

- Built retrieval-augmented generation (RAG) with LangChain, SentenceTransformers embeddings FAISS over a 10k+ document corpus; chunking, top-k retrieval and OpenAI API.
- Evaluated with RAGAS and LLM as a judge: retrieval precision @k, answer faithfulness and hallucination rate on a 50-question held out set – metrics and failure cases in README.

LoRA Fine-Tuning for Text Classification | 2026

- Fine-tuned DistilBERT with LoRA on ~50k wildfire and emergency domain social posts (3 class relevance labels); F1 0.89 on held out tests vs 0.76 zero shot baseline.
- Compared zero shot, full fine tune and LoRA (CUDA); exported weights for downstream RAG and capstone NLP pipeline – training curves and eval tables in README.

Agentic AI Assistant with Tool Calling | 2025

- Built an agentic AI assistant (LangChain) with tool/function calling – retrieval, SQL lookup and API tools – for multi-step wildfire intelligence and MLOps troubleshooting workflows.
- Added short term memory and structured tool routing for multi-step queries; README includes architecture diagram, example tool call traces and workflow screenshots.

Network Intrusion Detection (NIDS) | ML Paper | 2024

- Classification pipeline on NSL-KDD and CIC-IDS (Scikit-learn, XGBoost); Random Forest achieved~99% on held out NSL-KDD tests with false alarm vs detection rate analysis.